MEMORANDUM

What decides whether a flexible industrial load saves money

An independent analysis of Spain, South Australia and Minnesota, built entirely from published prices and published network tariffs.

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Evidence 25 years of hourly prices, three markets **Sources** all public, all cited

Executive summary

A factory that makes heat from electricity pays two separate bills. One is for the power itself. The other goes to the network company for delivering it, and that second bill is usually set by the single biggest burst of power the factory ever draws.

A thermal battery is built to win the first bill. It buys electricity when it is cheap, stores the energy as heat, and releases it steadily. But buying cheaply means buying in short hard bursts, which is precisely what the second bill punishes.

Across three markets and 25 years of published prices, it was always the second bill that decided whether a battery saved money or lost it.

And that bill turned on one detail: **which hours the network is watching when it looks for your biggest burst.** Some networks watch only certain hours, or charge far less for what they see outside them, and a battery can put its bursts where they are cheap. Other networks watch every hour of the year, so whenever the battery charges hard, that is the burst it pays for.

In Minnesota, where Otter Tail Power watches at any moment, a thermal battery lost money in all 36 combinations of location and year tested. In South Australia, where two network companies bill the same factory two different ways, the one watching a four hour evening window beat the one watching all year by 19 to 44 percentage points, in each of nine consecutive years.

The price of electricity mattered far less. Spain's average wholesale price ranged five-fold, from €34 per MWh in 2020 to €168 in 2022, and the saving in those two years barely moved: 17.1% and 16.8%. Across the same eight years it rose from **12.5% to 37.9%**, with one dip in 2021, driven by how often power went cheap rather than by how expensive it was on average.

Background and method

Everything here models one factory. It needs 10 MW of heat every hour of the year, and it has a thermal battery holding 12 hours of heat, which can be filled at up to four times its average draw. Every figure compares that factory against an identical one with no battery, buying power as it uses it. The gap between the two bills is what the battery is worth.

Every price used is a real published one, hour by hour: eight years of it for Spain, nine for South Australia, and eight for Minnesota, all ending in 2025. The network tariffs are published too, each taken straight from the operator's own schedule.

Tariffs are held at their 2025 values across every price year, so that only prices vary. The trends below are therefore statements about prices alone.

The battery follows a deliberately simple rule: when the store is about to run dry, go back and buy in the cheapest hour that could still be holding heat. That rule was checked against the exact optimum, solved as a linear program, and it comes out **1.5% worse and always in the same direction**. **Every saving reported here is therefore a floor, never a best case.** The charge rate of four times average draw is also not a free choice: below it the battery gives up real savings, and above it the extra connection needed stops paying for itself.

Three things are deliberately left out. What the battery costs to build. What it is worth to have heat available when the grid fails. And any income beyond buying power at better times.

> **So this measures what a battery saves on the electricity bill, and nothing else. It does not say whether building one pays for itself.**

Key findings

1 The billing rule decides the answer, and South Australia proves it on its own.

South Australia has two network companies. SA Power Networks looks for a factory's biggest burst only between 5pm and 9pm, and only from November to March. ElectraNet looks at any moment, all year round. A large factory could be billed by either.

So run the same battery, on the same prices, in the same years, and change nothing except which company sends the bill. Each figure is what the battery saved against the same factory without one.

Year	SA Power Networks watches a window	Para watches always	Brinkworth	Ardrossan	Berri	Gap vs best
2017	**+22.5%**	−2.0%	−7.5%	−12.3%	−14.6%	24.5
2018	**+27.2%**	+1.5%	−4.3%	−9.4%	−11.8%	25.7
2019	**+36.3%**	+9.9%	+4.0%	−1.3%	−3.8%	26.4
2020	**+39.0%**	−5.5%	−14.8%	−22.7%	−26.4%	44.5
2021	**+52.3%**	+10.3%	+1.3%	−6.5%	−10.2%	42.0
2022	**+45.6%**	+26.4%	+21.9%	+17.9%	+16.0%	19.2
2023	**+55.5%**	+23.9%	+16.8%	+10.5%	+7.5%	31.6
2024	**+54.2%**	+27.2%	+20.9%	+15.4%	+12.8%	27.0
2025	**+49.9%**	+20.7%	+13.9%	+8.0%	+5.2%	29.2

The four middle columns are ElectraNet, at each of the four places in South Australia a factory could physically connect.

Watching a window instead of watching always is worth **19 to 44 percentage points**, in every one of the nine years. Nothing else differs: same country, same prices, same weather, same battery, same currency. Connected to ElectraNet, the battery **loses money outright in 15 of the 36** year and location combinations.

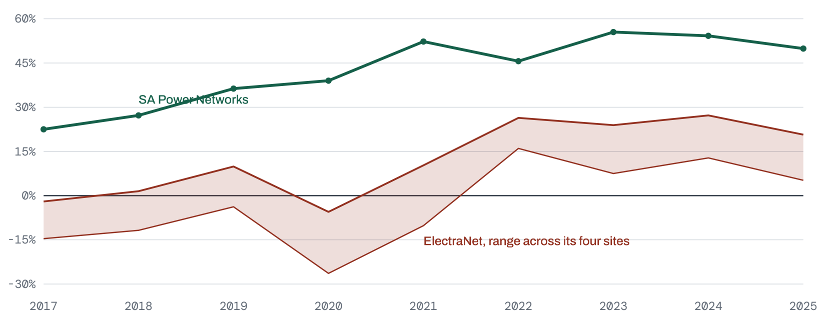

Figure 1. The same battery on two networks in the same state. The green line is SA Power Networks, which looks for your biggest burst only on summer evenings. The red band covers ElectraNet's four connection points, which look at any moment. The gap never closes.

2 There are only two moves available, and the tariff decides whether either one works.

A battery can do two things. It can chase cheap power, buying when the wholesale price is low. And it can stay out of the hours the network is watching. Running all four combinations on the same year separates what each is worth on its own.

	Chasing cheap power only	Staying out of watched hours only	Doing both	Extra from doing both
Spain	+33.9%	**-4.9%**	+38.7%	**+9.7**
South Australia	+39.6%	**+8.9%**	+49.8%	+1.3
Minnesota	**-59.1%**	-2.6%	-10.9%	+50.8

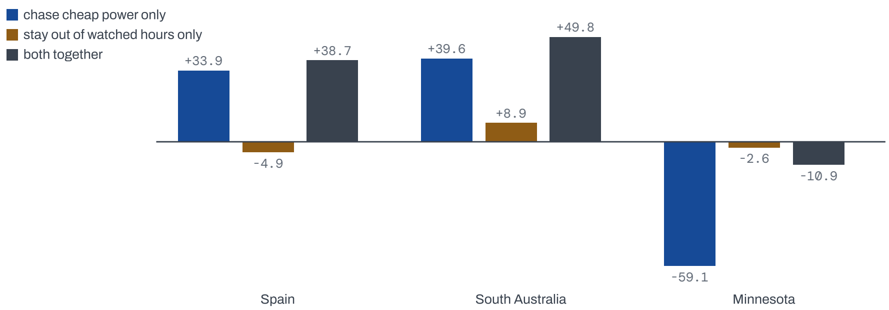

Figure 2. What each move earns alone, and what the two earn together. In Spain the grey bar is taller than the blue and orange added together, because the clock earns nothing on its own and only pays alongside cheap-power hunting. In Minnesota every bar is below the line.

Three different designs show up.

South Australia pays you to stay out. Avoiding 5pm to 9pm is worth 8.9% before a single cheap hour is chased. A factory needs no forecasting skill at all to benefit.

Spain does not pay you to stay out, but it stops you being punished. Staying in the cheap bands alone *loses* 4.9%. Its value is entirely in combination: chasing cheap power creates a big burst, and Spain's bands let you put that burst where it is charged almost nothing. That is why doing both is worth 9.7 points more than the two parts added together.

Minnesota punishes both moves. Chasing cheap power alone loses 59.1%, because the burst gets billed. There is no clock to play, so the second move is worth almost nothing. Both together still lose.

Minnesota also settles what is causing all of this, because the same utility offers two tariffs on the same grid, to the same factory, in the same years. Otter Tail Power's standard tariff bills whatever the meter recorded as the highest moment that month, and the battery loses money in every one of the eight years. Widened to nine MISO pricing points across four of them, it loses in all 36 combinations. Its Thermal Market Energy Pricing tariff bills a figure agreed in advance instead, and the same battery wins in all eight. **Nothing changed but one sentence in the rate schedule.** That tariff is already filed and approved.

3 Against gas, the case has improved fast, and 2025 is the first year it is clearly won.

A factory choosing today is choosing against a gas boiler, not an electric heater. Gas is cheap, so the comparison turns on what the factory pays for the carbon it emits.

Rather than argue about the right carbon price, the analysis works out the price at which the two come out equal. In Spain that number has collapsed.

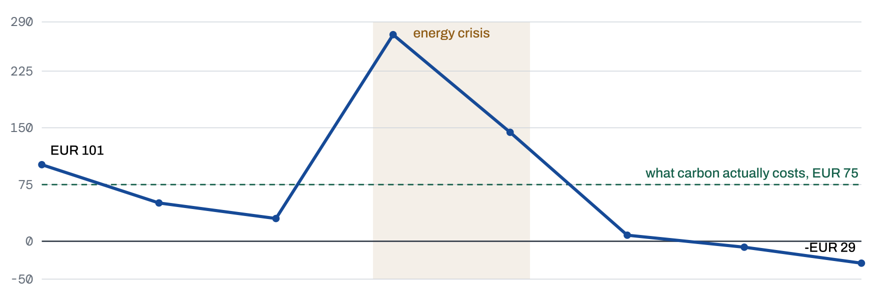

Figure 3. The carbon price at which heat from a battery costs the same as heat from gas, in Spain. Below the dashed line, the battery is already cheaper at the carbon price Europe actually charges.

Read it as: in 2018, carbon would have had to cost €101 a tonne before the battery beat gas. By 2025 the battery wins even if carbon costs nothing at all. Europe actually charges €75.

The two spikes are the European energy crisis, and they are worth understanding instead of removing. Gas more than doubled in 2022, which should have favoured the battery. It did the opposite, because the wholesale electricity the battery buys more than doubled too. Spain's average price ran €168 per MWh that year against €65 in 2025. **An energy crisis is not a one-way bet on storage, because both fuels are in it.**

The honest qualification is that 2025 is the best year in the series. Ignoring carbon and comparing fuel cost alone, the battery beat gas in only two of the eight Spanish years. **The trend is what to quote, not the single year.**

What would change this

Three assumptions could move these numbers. Each was tested by re-running the whole model across a range of values and recording how far the answer travels, so these are measured spreads, not estimates of uncertainty.

How far ahead an Australian operator can see

To buy heat now for use later, you have to know what later will cost. Spain and Minnesota both publish the next day's prices, so this is settled there. Australia has no such market: it publishes forecasts, but no price anyone can commit to. Six hours of useful sight was assumed.

Hours of sight	3	6	8	12	24
South Australian saving	not workable	**49.9%**	58.0%	69.7%	**73.1%**

Twenty-three points rest on that one assumption. Tested against the forecasts Australia actually published, the battery captured 37% of what perfect knowledge would give at six hours, and 80% at twelve. So the assumption is bounded by evidence without being settled by it.

What figure Otter Tail agrees to bill

TMEP bills demand on a number set in the service agreement, and the tariff itself does not say what that number is. Ten megawatts was assumed, the factory's steady load.

Agreed figure	10 MW	15 MW	20 MW	40 MW
Saving	**+20.6%**	+7.3%	**-6.0%**	-59.3%

Forty megawatts is the factory's actual metered peak, which is what it pays with no agreement at all. **The mechanism is the finding; the agreed number is where the value is really decided,** which is itself the useful point for anyone negotiating one.

South Australia is the best of the five Australian regions

Holding the same tariff fixed and swapping in each Australian region's prices:

	South Australia	Victoria	New South Wales	Queensland	Tasmania
Saving	**49.9%**	43.7%	42.5%	40.8%	**22.6%**

The mechanism holds everywhere. The size of the win does not, so South Australian figures should be read as the top of a range, not as a headline.

> Produced by `analyse_robustness.py`, `analyse_forecast_skill.py`, `analyse_tmep_baseline.py` and `analyse_nodes.py`. Every figure in this memo regenerates from published data by running the scripts in the repository.

Recommendation

A screening test that costs nothing

Before modelling a new market, read how its network measures peak demand. It is the single largest lever on the answer, and it is free to check.

Be careful how far that is pushed. It does not on its own decide whether a flexible load wins: ElectraNet measures at any moment and still pays from 2022 onward, in the years when prices swing hardest. What the reading reliably tells you is **how much of the saving survives the delivery bill**, and on identical prices that is worth 19 to 44 percentage points.

Where prices do matter, what matters is not their level but how far the cheapest hours sit below the daily average as a share of it. Tested across five Australian regions, that ranked them correctly where the more obvious measure, the gap between the day's highest and lowest price, ranked them wrong.

One thing to check alongside it. A charge on peak demand is a fixed cost, so it falls hardest on plants that do not run flat out. Tested against a two-week maintenance shutdown, a five-day week and a day-shift pattern, the saving held everywhere, but **at 48% utilisation the same capacity charge costs 2.1 times more per unit of heat**, in all three markets. Any load that runs intermittently is being taxed for reasons unrelated to what it costs the network to serve.

Treat TMEP as a template

It is filed, approved and already operating. Any utility serving a large flexible load could write the same sentence into its own schedule, and the finding above puts a number on what that sentence is worth.

The gap worth arguing about is duration

Nothing in any of these three markets pays for storage depth beyond about four hours. Buying at better times pays up to the day-ahead horizon and exactly nothing after it. MISO's capacity auction pays $217 per MW-day but asks for only four consecutive hours, so a hundred-hour battery earns what a four-hour one earns. Australia has no capacity market at all. Riding through the worst stretch of a year takes 99 to 164 hours.

> **Everything between 24 and 164 hours is unpaid by every market examined.** The market cannot currently buy the thing long-duration storage exists to provide.

What would take this further

Three things, none of them confidential. What it costs a customer when heat stops, which is the missing term in every firmness calculation here. What baseline demand is realistically achievable in a TMEP-style negotiation, since that is where the value is decided. And whether the right reference point is a gas boiler or an inflexible electric one, because this study reports both and they answer different questions.

Prices. OMIE for Spain, AEMO for the Australian regions, MISO market reports and monthly archives for Minnesota. Hourly throughout, 2017 to 2025.
Tariffs. CNMC and the BOE for Spain, SA Power Networks and ElectraNet price schedules, Otter Tail Power's Minnesota rate summary and riders. Every rate cited inline in the code.
Gas and carbon. Eurostat, the AER's STTM register, and the EIA. EU ETS and Australia's Safeguard prescribed unit price.

Model. Every tariff here is also available as a working model, which takes a rate schedule and returns whether a flexible load wins or loses on it, and which part of the bill decided.

44 decisions logged, 18 of them judgment calls. 19 claims made during the work and later overturned, kept on the record.